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B.TECH. III SEMESTER MAIN/BACK EXAMINATION 2024-25 COMPUTER SCIENCE AND ENGINEERING 3CS1–01 -

Advanced Engineering Mathematics

3AD1-01, 3AM1-01, 3AI1-01, 3CM1-01, 3CY1-01, 3CD1-01, 3CS1-01, 3IT1-01, 3IC1-01 - Common to CS, IT, AI, DS, MC, CM, CD, CA, AD, AM, CY, IO

Time : 3 Hours] **[Max. Marks :** 70

[Min. Passing Marks :

Instructions to Candidates :

Part–A : Short Answer Type Questions (up to **25** words) 10 × 2 = 20 marks. All 10 questions are compulsory.

Part–B : Analytical/Problem Solving questions 5 × 4 = 20 marks. Candidates have to answer 5 questions out of 7.

Part–C : Descriptive/Analytical/Problem Solving questions 3 × 10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of the following supporting materials is permitted during examination. (Mentioned in form no. 205).

1_____

2_____

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Part–A

1. Write down the Gauss forward interpolation formula. 2
2. State the necessary and sufficient conditions for the maximum of a function $f(x)$. 2
3. Evaluate $\Delta \tan^{-1}(x)$. 2
4. Define objective function. 2
5. Define continuous random variable. 2
6. Write down the definition of skewness. 2
7. Find the Hessian matrix of the function $f(x) = x^2 - y^2$. 2
8. Write the formula for the Simpson's 1/3 rule. 2
9. What is a random variable ? 2
10. Find the expected value of the data : 2

X	P(x)
4	1/12
5	1/12
6	1/4
7	1/4
8	1/6
9	1/6

Part–B

1. Find the missing value of the following table : 4

X	Y
45	3.0
50	–
55	2.0
60	–
65	–2.4

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2. Write the dual to the primal

Minimize :

$$z = 10x_1 + 8x_2$$

Subject to

$$x_1 + 2x_2 \geq 5,$$

$$2x_1 - x_2 \geq 12,$$

$$x_1 + 3x_2 \geq 4$$

$x_1 \geq 0$, and x_2 unrestricted. 4

3. Find the line of best fit relating to y to x for the following data, plot the line and the data points : 4

X	Y
2	3.0
3	5.0
4	5.5
5	6.0
6	8.0
7	9.5

4. A five-figure number is formed by the numbers 0, 1, 2, 3, 4 without repetition. Find the probability that number formed is divisible by 4. 4

5. Find the maximum and minimum values of the function 4

$$f(x) = 12x^5 - 45x^4 + 40x^3 + 5$$

6. State the Kuhn-Tucker conditions. 4

7. A sample of 100 items is taken from a batch known to contain 40% defectives. Using normal approximations find the probability that contains :

A. At least 44 defectives,

B. Exactly 44 defectives. 4

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Part-C

1. Find the second degree polynomial that passes through the points : 10

$$(1, -1), (2, -1), (3, 1), (4, 5)$$

2. Find the optimal solution of the following transportation problem : 10

		To 1	2	3	4	Supply
From	1	9	16	15	9	15
	2	2	1	3	5	25
	3	6	4	7	3	20
	Demand	10	15	25	10	

3. Calculate the Spearman's rank correlation coefficient for the following data : 10

X	y
65	68
63	66
67	68
64	65
68	69
62	66
70	68
66	65
68	71
67	67
69	68
71	70

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4. The joint density function of X and Y is given by : 10

$$f(x,y) = \begin{cases} 2e^{-x}e^{-2y} & , \quad 0 < x < \infty, 0 < y < \infty \\ 0 & , \quad \text{otherwise} \end{cases}$$

Find :

(a) $P\{X > 1, Y < 1\}$

(b) $P\{X < Y\}$

(c) $P\{x < a\}$

5. Find the dimension of a cylindrical tin (with top and bottom) made up of a sheet metal to maximize its volume such that the total surface area is equal to $A_0 = 24\pi$. 10

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